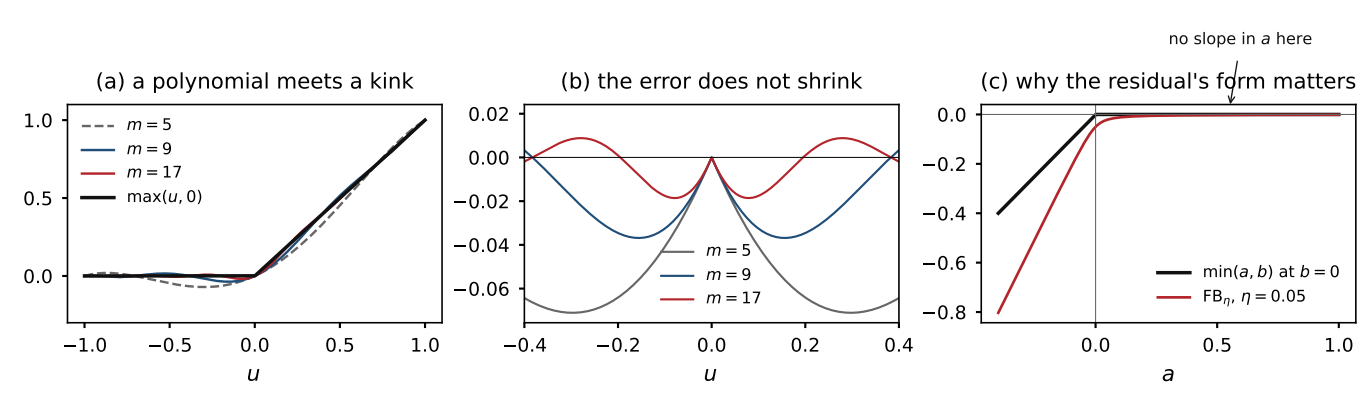
(a) a polynomial meets a kink
$m = 5$
$m = 9$
$m = 17$
$\max(u, 0)$
1.0
0.5
0.0
−1.0
−0.5
0.0
0.5
1.0
u
(b) the error does not shrink
0.02
0.00
−0.02
−0.04
−0.06
$m = 5$
$m = 9$
$m = 17$
−0.4
−0.2
0.0
0.2
0.4
u
no slope in a here
(c) why the residual's form matters
0.0
−0.2
−0.4
−0.6
−0.8
$\min(a, b)$ at $b = 0$
FB_η, $\eta = 0.05$
0.0
0.5
1.0
a